LOGOS: Benchmarks for DDA Peptide Search

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Abstract

LOGOS is a proteomics search engine written in Rust with a CUDA backend. It reads Thermo RAW files directly and searches on one CPU node or one GPU. Against Sage 0.14.7 on the same node, chemistry and 22 cores, it runs closed DDA 12.2× and open search 6.6× faster on the CPU, and open search 16.6× faster on one GPU, while reporting more peptide sequences at an entrapment-measured error. At scale it searches the 79.7 GB HEK293 deposit PXD001468 in an open search in 12.13 s on one GPU, and one gut metaproteome against 111.5 million sequences in 20.3 s on an L40S (recorded on an earlier build). CPU and GPU builds produce byte-identical identifications. The record behind this report covers 216 RAW files from 11 public datasets and 172 recorded comparisons against external tools, released with it as data. Measurements were taken through 31 October 2026 on the Warrior cluster at Wayne State University; this report covers data-dependent acquisition only.

1 Across datasets

Figure 1 sets LOGOS against the strongest tool or published result available on each benchmark: Sage where Sage was run on the same files and node, the authors' deposited search where that is the reference, and CHIMERYS, a chimeric-spectrum engine, where its deposits exist. Counts are distinct sequences or PSMs at 1% FDR under one counting rule; a filled marker means LOGOS's count is the entrapment-validated one. Table 1 gives the numbers behind every bar.

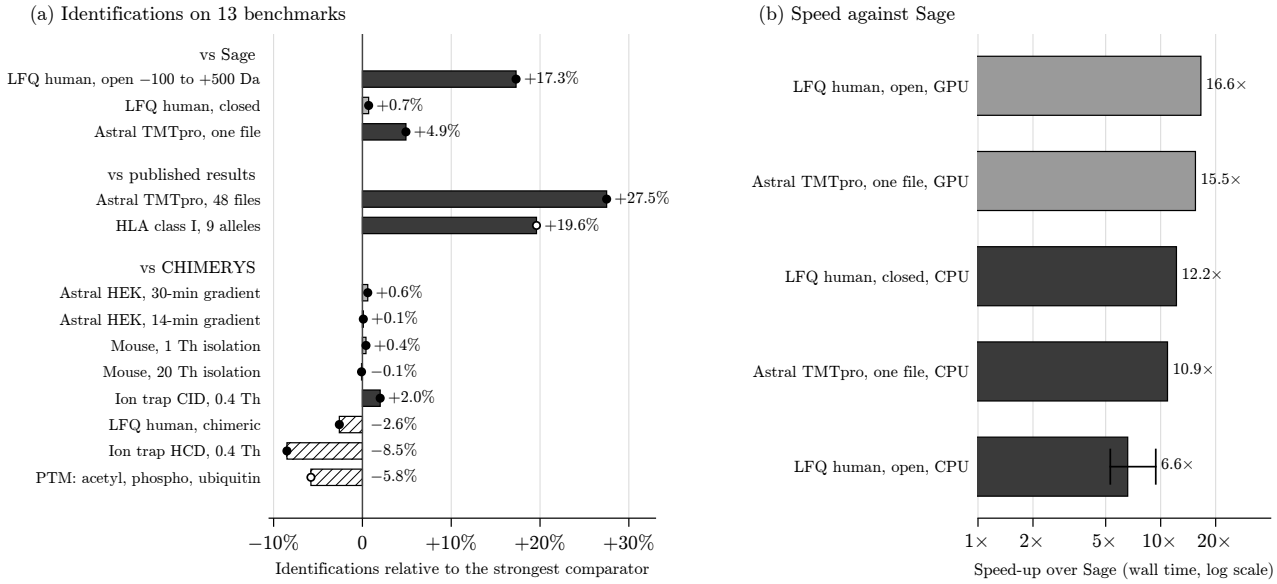


Figure 1: (a) LOGOS identifications relative to the strongest comparator on 13 benchmarks across 6 public datasets, 4 instrument families and 9 search tasks. Dark bars: ahead by more than 1%; light: within 1%; hatched: behind. Filled marker: entrapment-validated LOGOS count; open: target-decoy only. (b) Wall-time speed-up over Sage; the whisker spans the three nodes measured for open search (Section 4). Rows marked GPU compare one GPU with Sage on CPU.

Reading the scoreboard. Against Sage LOGOS reports more sequences on every dataset tested: 0.7% more in closed search at equal measured error, 4.9% more on Astral TMTpro and 17.3% more in open search, while running 12.2× to 16.6× faster. Against the authors' own searches it reports 27.5% more PSMs on a 48-file TMTpro study, ahead in all 48 files, and 19.6% more peptides on HLA immunopeptidomics. Against CHIMERYS it ties within 1% on Astral gradients of 14 and 30 minutes and on wide-window data at 1 and 20 Th, edges ahead on ion-trap CID, and trails on ion-trap HCD, PTM sets and chimeric LFQ.

Table 1: Scoreboard: LOGOS against the strongest comparator on each benchmark. Status gives the date of the measurement; rows from 31 October were rerun on the current build.

Benchmark	Data	Comparator	LOGOS	Comparator	Δ	Error control	Status
LFQ human, open −100 to +500 Da	PXD028735 Q Exactive HF-X	Sage 0.14.7	35,595	30,343	**+17.3%**	entrapment 1.04% vs 0.90%	measured 2026-10-31
LFQ human, closed	PXD028735 Q Exactive HF-X	Sage 0.14.7	31,718	31,489	+0.7%	entrapment 1.24% each	measured 2026-10-30
Astral TMTpro, one file	PXD058160 Orbitrap Astral	Sage 0.14.7	7,568	7,212	**+4.9%**	entrapment 0.76% vs 1.02%	measured 2026-10-31
Astral TMTpro, 48 files	PXD058160 Orbitrap Astral	Authors' Comet + LDA	1,700,508	1,333,804	**+27.5%**	entrapment 0.997%; ahead in 48 of 48 files	recorded 2026-10-22
HLA class I, 9 alleles	MSV000084172 Orbitrap	Deposited Spectrum Mill	20,625	17,248	**+19.6%**	target-decoy; reference without matched FDR	recorded 2026-09-19
Astral HEK, 30-min gradient	PXD046453 Orbitrap Astral	CHIMERYS (recounted)	91,865	91,362	+0.6%	LOGOS entrapment 1.02%	recorded 2026-09-29
Astral HEK, 14-min gradient	PXD046453 Orbitrap Astral	CHIMERYS (recounted)	55,383	55,309	+0.1%	LOGOS entrapment 1.03%	recorded 2026-09-29
Mouse, 1 Th isolation	PXD053241 Q Exactive HF-X	CHIMERYS (recounted)	68,896	68,624	+0.4%	LOGOS entrapment 1.04%	recorded 2026-09-29
Mouse, 20 Th isolation	PXD053241 Q Exactive HF-X	CHIMERYS (recounted)	39,971	40,007	-0.1%	LOGOS entrapment 1.05%	recorded 2026-09-24
Ion trap CID, 0.4 Th	PXD053241 Orbitrap Eclipse	CHIMERYS (recounted)	21,364	20,941	**+2.0%**	LOGOS entrapment 1.00%; CHIMERYS's database	recorded 2026-09-29
LFQ human, chimeric	PXD053241 Q Exactive HF-X	CHIMERYS (recounted)	59,606	61,223	-2.6%	LOGOS entrapment 1.05%	recorded 2026-09-23
Ion trap HCD, 0.4 Th	PXD053241 Orbitrap Eclipse	CHIMERYS (recounted)	46,340	50,624	-8.5%	LOGOS entrapment 0.98%	recorded 2026-09-29
PTM: acetyl, phospho, ubiquitin	PXD053241, PXD013615 Q Exactive HF-X	CHIMERYS (recounted)	41,985	44,560	-5.8%	target-decoy	recorded 2026-09-23

2 Speed and scale

Figure 2 collects every speed comparison in the record in which LOGOS and another tool did the same work on the same input. The search rows are the Sage comparisons of Section 4. Rescoring is 11–34× faster than C++ Percolator on the same input, and the MSBooster features are 75× faster with byte-identical output. LOGOS searches a RAW file to identifications 94× faster than ThermoRawFileParser converts that file to the mzML other engines read.

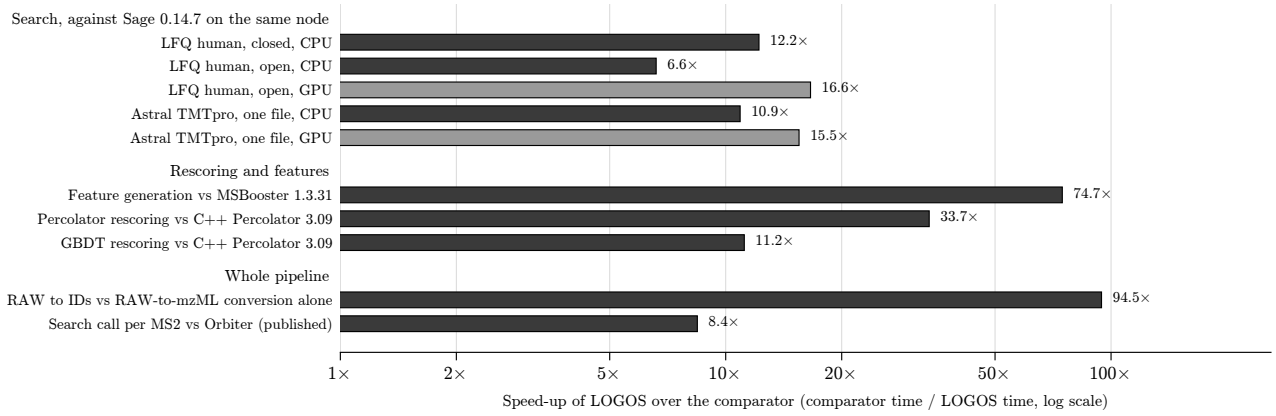


Figure 2: Speed-up of LOGOS over each comparator, as the comparator's time divided by LOGOS's, on the same input and work. Search: wall time against Sage 0.14.7 on CPU on the same node, LOGOS on CPU (dark) or on one GPU (light). Rescoring and features: CPU, same PIN input; MSBooster's 58.29 s includes about 26 s of Java start-up. Whole pipeline: LOGOS's full search of one Astral file from RAW against ThermoRawFileParser converting that file alone; LOGOS's median search call per MS2 (search only, ±50 ppm precursor window, CPU, 4,096 MS2 of PXD017823) against Orbiter's published 17 ms median, on different hardware.

Where no other engine has been run on the same input, Table 2 reports absolute throughput. LOGOS searches the 24-file HEK293 deposit of Chick et al. (PXD001468, 79.70 GB) in an open search in 12.13 s on one GPU, reporting 132,592 sequences at an entrapment-measured 1.03%. It searches one gut-metaproteome run against the 111.5 million sequences of ComPIL 2.0, scoring 178.9 billion candidates, in 20.26 s on an L40S as recorded on 18 September; the current build takes 65.05 s on an RTX PRO 4500 with identical identifications.

The cascade asks what share of every MS2 spectrum in PXD001468 can be explained at controlled error.

Table 2: Throughput at scale and latency in real time. Inputs staged in RAM; GPU counts are physical devices. Rows were recorded on earlier builds before 31 October 2026 and not rerun on the current build.

Workload	Result	Identifications	Hardware	Note
HEK293 open search, PXD001468: 24 RAW, 79.70 GB, 1,121,149 MS2	12.13 s; 394 GB/min; 92,428 MS2/s	132,592 sequences, 688,428 PSMs at 1%; entrapment FDP 1.03%	1x H200 NVL, 32 EPYC 9554 cores	median of 3; RAWs in RAM
Same files, closed search	7.89 s; 606 GB/min	–	1x H200 NVL	RAWs in RAM
Same open search on a workstation GPU	20.30 s; 236 GB/min	–	1x RTX PRO 4000, 32 EPYC 9454 cores	RAWs in RAM
HEK293 cascade: every MS2 through nine layers	939,455 of 1,121,149 MS2 explained (83.8%)	entrapment FDP 0.87% overall (Ne 4,025)	1x L40S	two layers above 1% alone; without them 81.96%
ComPIL 2.0 metaproteome: one gut RAW against 111,478,540 sequences (84 GB FASTA, 64.06 G residues)	20.26 s; 178.9 billion candidates scored	38,649 PSMs, 10,326 peptides, 8,077 proteins at 1% (target-decoy)	1x L40S, 32 EPYC 9554 cores	recorded 2026-09-18, median of 2; performance settings; no entrapment arm
ComPIL 2.0, same search on the current build	65.05 s (repeats 65.05, 65.11, 64.96); same 178.9 billion candidates	identical: 38,649 PSMs, 10,326 peptides, 8,077 proteins	1x NVIDIA RTX PRO 4500 Blackwell, 32 EPYC 9454 cores	run 2026-10-30-compil2; GPU-bound: the scan is 62.9 s of the run
ComPIL 2.0, published-protocol settings	80.2% of the 7,565 published H2 peptides recovered	after I/L normalisation	Apple GPU (Metal)	separate run from the speed run
Astral TMTpro, PXD058160: 15 RAW, 39.5 GB	20.91 s; 113 GB/min	392,305 PSMs, 97,666 sequences at 1%	1x RTX PRO 4000	median of 3; one process

Each layer searches only the spectra that every earlier layer left unexplained. One learner scores the rows of all layers, and target-decoy competition runs per spectrum across layers. The error is measured on a separate entrapment arm that runs the whole layered pipeline. Figure 3 shows the result: 939,455 of 1,121,149 spectra (83.8%) explained at an overall entrapment FDP of 0.87%.

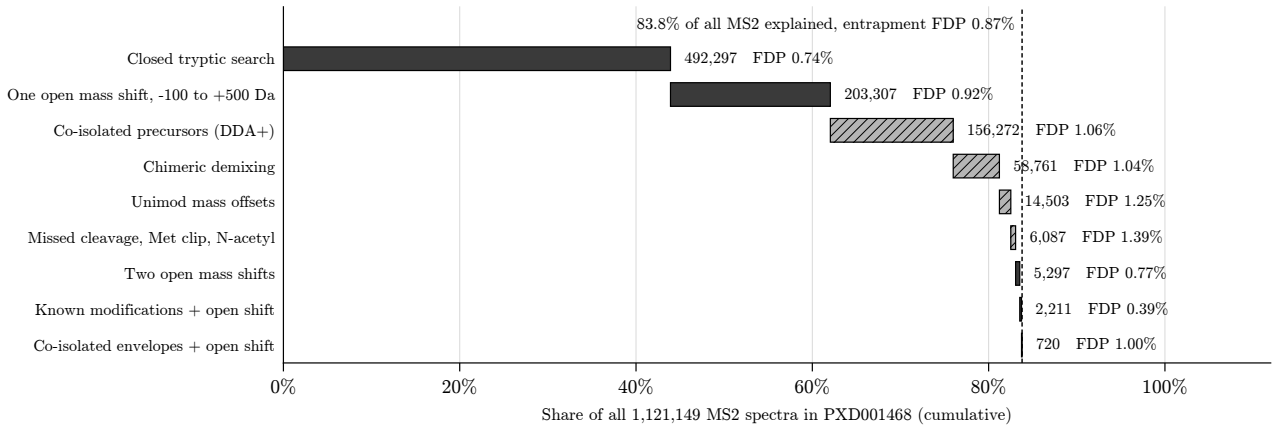


Figure 3: The HEK293 cascade on all 24 files of PXD001468. Each bar is the share of all MS2 that a layer explains, placed after the layers before it; the label gives spectra explained and that layer's own entrapment FDP. Hatched layers run above 1% on their own (1.04–1.39%); without the two highest, Unimod offsets and missed cleavage, the cascade explains 918,865 spectra (82.0%). Of the spectra left unexplained, about half are chimeric and a third carry too little information for any engine; well-formed spectra that stay unexplained are 3.3% of the total (6-file subset).

3 Real-time acquisition decisions

In real-time search the instrument asks, for every MS2 it records, whether to spend an MS3 scan on that precursor. LOGOS answers with one pipeline per MS2: candidate search, XCorr, a lookup of predicted fragment spectra, a causal retention-time check, a frozen score model and a trigger policy. The measurements below replay recorded acquisitions of PXD043660 (Thermo real-time search on Orbitrap Ascend and Eclipse) and PXD058160 (Orbitrap Astral) one MS2 at a time, at batch 1, with RAW decoding excluded; every percentile is recomputed from the replay's raw per-MS2 samples by `eval/latency.py`.

On the TKO file a decision takes a median 0.884 ms and 1.205 ms at the 99th percentile on one RTX PRO 4500, over three interleaved runs of 27,674 decisions, every one of which equals the offline reference. The

previous binary measured 0.925 ms in the same job. One of the 83,022 timed decisions exceeded the 80 ms budget Thermo's implementation sets, a single 11.3 s host stall; on the HYpro file 19 of 96,600 did. On the Astral file, whose recorded MS2 arrive a median 24.8 ms apart, the median decision takes 1.98 ms and none of 135,711 exceeds 80 ms, from a single run.

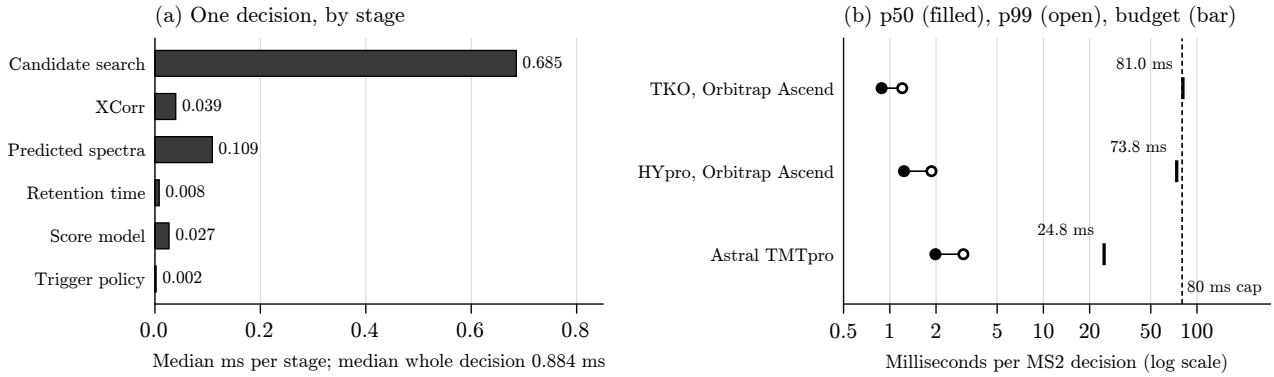


Figure 4: (a) The median time of each stage of a TKO decision; the stage medians sum to 0.87 ms, slightly less than the median whole decision, as medians do. (b) Each replay's median (filled) and 99th-percentile (open) decision time against its own budget (bar): the median time to the next MS2 for the Orbitrap Ascend files, the median recorded MS2 gap for the Astral file. The dashed line is the 80 ms cap of Thermo's real-time search.

Table 3: Decision latency, recomputed from raw per-MS2 samples (runs `2026-10-22-rts-latency`, `2026-10-22-astral-latency`; job 40703237, RTX PRO 4500, EPYC 9454). Times in ms; over-80 ms counts are over all runs.

Replay	Binary	Runs	p50	p99	Over 80 ms	Matching the offline reference	Budget
TKO, Orbitrap Ascend (27,674 MS2)	s7	3	0.884	1.205	1 of 83,022	27,674 of 27,674	81.0
HYpro, Orbitrap Ascend (32,200 MS2)	s7	3	1.237	1.868	19 of 96,600	32,200 of 32,200	73.8
Astral TMTpro (135,711 MS2)	astral-opt	1	1.983	3.015	0 of 135,711	135,704 exact of 135,711	24.8

Speed matters only if the decisions are good. Replaying all 24 PXD043660 acquisitions with the authors' own database, and trigger rules chosen on files other than the one being replayed, LOGOS triggers on more of the authors' passing scans with fewer MS3 scans, and the scans it triggers carry an entrapment-measured FDP of 0.56% where Thermo's recorded triggers carry 2.95%, lower in all 24 files (sign test $p = 1.2 \times 10^{-7}$). It does not win everywhere: raw protein counts tie at 5,442, which the rule we set ourselves counts as a loss, both HYpro regimes trail by up to 10 proteins, and LOGOS has more proteins in only 6 of the 24 files. This replay and the latency runs use different recipes (the authors' database and v11 rules here; an earlier database and v8 rules for the timing), so neither number is claimed for the other.

Table 4: Replay of 24 Thermo real-time-search acquisitions (PXD043660), LOGOS / Thermo recorded; bold where LOGOS is higher. Entrapment FDP in percent, of the triggered scans' identifications, both masks classified by the same scorer. Recorded 21 October 2026 (run `2026-10-21-rts-replay`).

Regime	Author-pass scans	Modified peptides	Raw proteins	MS3 scans	Entrapment FDP
HYpro, Ascend (2 files)	**15,954** / 15,950	**9,542** / 9,390	2,322 / 2,332	20,036 / 20,041	**1.79** / 6.97
HYpro, Eclipse (2 files)	**17,408** / 17,194	**10,581** / 10,473	2,673 / 2,675	20,333 / 20,564	**1.48** / 6.20
TKO11, Ascend (10 files)	**67,328** / 67,176	**16,259** / 15,973	**2,528** / 2,518	73,561 / 73,645	**0.27** / 1.75
TKO11, Eclipse (10 files)	**67,024** / 66,693	**16,545** / 16,229	**2,665** / 2,655	72,999 / 73,202	**0.24** / 2.21
Pooled, 24 files	**167,714** / 167,013	**30,766** / 30,182	5,442 / 5,442	186,929 / 187,452	**0.56** / 2.95

On the 12 Orbiter acquisitions of PXD017823 the trigger rules were chosen retrospectively and the database differs from the authors', so these results are weaker evidence. One rule triggers 38,169 of the authors' passing scans against Orbiter's 30,815, with 40,340 MS3 scans against 41,647, at an estimated 2.7% false-trigger rate against 16.4%, but reaches fewer raw protein accessions (8,857 against 9,245). Another reaches 8,710 proteins at 1% picked-protein FDR against 8,315, 6,563 of them with two or more peptides against 5,821, using 39,536 MS3 scans. A separate measurement times the search call alone with a broad closed precursor window of ±500 Da: 32.7 ms per MS2 at the median on an NVIDIA A30, against 2.74 s on CPU.

4 Head to head with Sage on the current build

Both engines ran in the same Slurm job on the same node, interleaved with the order rotated between repeats; times are medians of three or more wall-clock measurements from input file to identifications, container start included. Sage's speed varies by a factor of 2.8 between these nodes, so every ratio is taken within one node.

Table 5: Wall time per search (seconds, median) and speed relative to Sage on the same node.

		Wall time (s)			Speed-up			
Benchmark	Node	CPU	GPU	Sage	CPU	GPU	Binary	Slurm job
Closed DDA, LFQ human	ymh4 · EPYC 9454	1.41	1.53	17.25	12.2	11.3	47068225	40931267
Open −100 to +500 Da, LFQ human	ymh4 · EPYC 9454	6.96	2.79	46.23	6.6	16.6	47068225	40928343
Open −100 to +500 Da, LFQ human	bbc1 · EPYC 9374F	7.08	–	37.79	5.3	–	47068225	40928337
Open −100 to +500 Da, LFQ human	arw3 · Gold 6326	11.36	–	106.4	9.4	–	6671172e	40928265
Astral TMTpro, one file	ymh4 · EPYC 9454	3.68	2.58	40.06	10.9	15.5	47068225	40931267

Identification counts use one rule for both engines: Sage's results are converted to the same input format and counted by the same program, the open counter `eval/count.py`, which gives exactly the engine's integers on every counted run (run `2026-11-01-eval-parity`). The entrapment arm adds a second database, equal in size and absent from the sample, to each search; every accepted hit to it is a known false discovery, and $\mathrm{FDP} = N_e(1 + 1/r)/(N_t + N_e)$, with the ratio r of entrapment to target sequence space computed from each search's own digestion (1.0606 for the LFQ human database, 0.9998 for the Astral mouse database).

Table 6: Identifications at 1% FDR on the target database alone and in the entrapment arm, at sequence and PSM level. FDP in percent.

	Target only		Entrapment arm, sequences			Entrapment arm, PSMs		
Engine	Sequences	PSMs	N_t	N_e	FDP	N_t	N_e	FDP
Closed DDA, LFQ human								
LOGOS	33,102	59,769	31,718	204	1.24	57,154	349	1.18
Sage 0.14.7	32,453	58,668	31,489	202	1.24	56,594	337	1.15
Open −100 to +500 Da, LFQ human								
LOGOS	34,130	70,886	32,086	166	1.00	65,824	353	1.04
LOGOS + LDA rescoring	–	–	35,650	203	1.10	73,449	390	1.03
LOGOS + calibration + LDA	37,178	77,433	35,595	192	1.04	73,438	379	1.00
Sage 0.14.7	31,744	73,148	30,343	141	0.90	68,848	359	1.01
Astral TMTpro, one file								
LOGOS	7,769	24,167	7,568	29	0.76	23,364	124	1.06
Sage 0.14.7	7,470	22,409	7,212	37	1.02	21,649	94	0.86

LDA rescoring alone raises the open-search error to 1.10%; with mass calibration it holds 1.04%, the configuration reported. On Astral TMTpro LOGOS reports more sequences at lower error, while at the PSM level Sage is the more conservative.

5 Method and correctness

- **Hardware.** Warrior compute nodes at Wayne State University, shared with other tenants: ymh4 (AMD EPYC 9454, NVIDIA RTX PRO 4500 Blackwell), bbc1 (AMD EPYC 9374F), arw3 (Intel Xeon Gold 6326); 22 cores per job, inputs staged in RAM.
- **Chemistry.** LFQ human: trypsin, one missed cleavage, 7–30 residues, carbamidomethyl C, up to two oxidised M, 10 ppm precursor (open: −100 to +500 Da), 20 ppm fragments, charges 2–4, deisotoping. Astral TMTpro: TMTpro on K and the N-terminus, two missed cleavages, 7–50 residues, 50 ppm precursor, 10 ppm fragments, charges 2–6. Sage 0.14.7 release defaults with this chemistry, including its LDA rescoring and retention-time prediction.
- **Backend parity and determinism.** CPU and GPU builds produce byte-identical identification files (open search: the same hash, 583ae51be609…, on every run); the same binary, input and settings give the same output on every run, because reductions are order-independent.

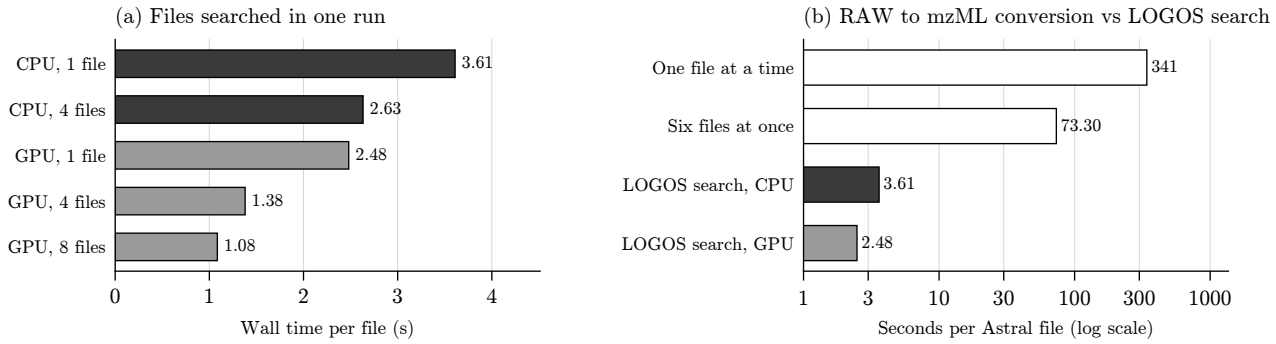


Figure 5: (a) Wall time per file when several Astral TMTpro files are searched in one run; each file's output is byte-identical to that file searched alone (jobs 40912674, 40912675, 40911945). (b) ThermoRawFileParser 2.0.0-dev converting one 2.6 GB Astral file to the mzML that Sage reads, alone and six at once (job 40913997), against LOGOS's whole search of that file from RAW.

- **The CPU path is the reference.** GPU kernels are tested against it, and new CPU code against brute-force implementations, mutation-checked so a broken variant fails its test.
- **Evidence.** Every number measured on the current build has a run record (`runs/`) naming the node, the binary and its SHA-256, the SHA-256 of every input, every repeat's wall time and the hash of every output count; the speed and identification tables of this report are generated from those records and a check fails the build if they differ. Inputs are pinned to their public deposits by SHA-256, the entrapment database is regenerated bit-for-bit by its generator, and the counting rule is open code with tests.
- **What LOGOS takes from Sage.** The CPU open-search preselection uses Sage's fragment-index layout (mass-sorted buckets of 8,192 peptides, fragments sorted within each); Sage's LFQ code was the design reference for LOGOS's MS1 quantification. On Astral TMTpro Sage spends 23.3 of its 39.4 s reading mzML at about four busy cores, so LOGOS's lead there comes mostly from reading RAW and preparing the database rather than the search core alone.

6 Limitations

LOGOS trails CHIMERYS on ion-trap HCD (−8.5%), on three PTM sets (−5.8% combined) and on chimeric LFQ (−2.6%), and quantifies half as many precursors as CHIMERYS on the LFQbench mixtures at equal accuracy (17,762 against 35,053; median error 0.175 against 0.177). Rows dated before 31 October come from earlier builds and are reported as recorded; several compare against a recount of the reference's deposited results rather than a fresh run. Speed ratios depend on the node, and every node was shared with other jobs. The complete record behind this report, 172 comparisons with their caveats and notebook sources, is in `reports/data/catalog.json`.

7 Scope of the record

Table 7 lists the public datasets behind the benchmark record and the tools each was compared against.

Table 7: Public datasets in the benchmark record. Rows counts the recorded measurements that cite the accession.

Accession	Dataset	Instrument	RAW	Benchmarked	Compared against	Rows
PXD001468	HEK293 proteome, Chick et al. 2015	Q Exactive	24	Open search at scale, closed search, explaining every spectrum in layers	Published counts	154
PXD058160	Astral TMTpro, four mouse tissues	Orbitrap Astral	48	TMT identification depth, speed vs Sage, multi-file throughput	Authors' Comet + LDA deposit; Sage 0.14.7	31
PXD043660 + PXD023203 + PXD017803	Thermo real-time search runs, TMT SPS-MS3	Orbitrap Eclipse and Ascend	32	Real-time acquisition decisions replayed, error rate of triggered MS3	Thermo's recorded real-time search triggers	73
PXD017823	Real-time search TMT study	Orbitrap	24	Identification depth and protein-level error	Authors' deposited passing scans	42
PXD058918	Astral TMTpro18 fractions	Orbitrap Astral	24	TMTpro18 identification depth	Authors' deposited search	9
PXD046453	Astral DDA, HEK 1 μg, 14/15/30-min gradients	Orbitrap Astral	15	Chimeric DDA+ depth on Astral 14- and 30-minute gradients	CHIMERYS (recounted)	107

Table 7, continued

Accession	Dataset	Instrument	RAW	Benchmarked	Compared against	Rows
PXD053241 (+ PXD023218, PXD013615)	CHIMERYS benchmark deposits: LFQ human, wide-window mouse, ion trap, PTM	Q Exactive HF-X, Orbitrap Eclipse	21	Wide-window chimeric DDA (8 widths), ion-trap HCD and CID (9), PTM sets (3), LFQ human	CHIMERYS, Sequest HT, Comet, MS-GF+, MetaMorpheus deposits	72
PXD028735	LFQbench three-species mixtures and LFQ human	Q Exactive HF-X	7	Closed and open search speed and depth vs Sage; quantification accuracy and depth	Sage 0.14.7, CHIMERYS, ProteoBench submissions	31
MSV000084172	HLA immunopeptidomics, 9 alleles	Orbitrap	18	Non-specific peptide identification	Deposited Spectrum Mill results	4
PXD022433 / MSV000082943	ComPIL 2.0 gut metaproteome	Orbitrap	1	Search against 111.5 M protein sequences	Published ProLuCID / DTASelect peptides	15
PXD012986, PXD000612	Mouse tissue; HeLa phospho	Orbitrap	2	Rescoring models	Percolator	10
Total			216			